Pre-work · Module 5 Integration

All ten Part A chapters. This is a synthesis sheet, not a new chapter. Nothing on it is material you have not already read, and that is the point: everything here crosses at least two chapters, so it is the sheet that shows you what you organized separately and never joined together. Work it with the notes closed and open them only to correct.

Module 5 · 5 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Rebuild Module 5 as eight visuals that each cross more than one chapter. Nothing here belongs to a single sheet, so if a chunk feels easy you are probably answering it from one chapter only.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **HUB**

The whole nervous system on one page

The CNS at the hub, brain and cord drawn in outline. Spoke out to the twelve cranial nerves, the thirty one spinal nerves, the five plexuses, the ganglia, the sensory receptors, and the three kinds of effector. Write the count beside every spoke that has one. If a count is missing you have found a gap.

2 **CHAIN**

One stimulus, all the way through

One long chain with arrows. Receptor in the skin, sensory neuron, posterior root and its ganglion, posterior horn, ascending tract, decussation, thalamus, somatosensory cortex, then motor cortex, internal capsule, cerebral peduncle, pyramids, anterior horn, anterior root, spinal nerve, skeletal muscle. Mark the two places the pathway crosses the midline.

3 **SPLIT**

CNS against PNS, every term that changes

One split. Cluster of cell bodies, bundle of axons, myelinating cell, neurolemma present or absent, regeneration possible or not, and whether gray matter is inside or outside. Six rows on each branch, and every row is a pair you have already met on a different sheet.

4 **GRID**

Where every cell body sits

Neuron down the side: somatic sensory, somatic motor, interneuron, sympathetic preganglionic, sympathetic postganglionic, parasympathetic preganglionic, parasympathetic postganglionic. Location of the cell body, and CNS or PNS across the top. This grid is the single hardest thing in the module to hold, and it is worth the frame.

5 **LADDER**

From the scalp inward to the neuron

Rungs from the outside in: bone, periosteal dura, meningeal dura, arachnoid, subarachnoid space with CSF, pia, cerebral cortex as gray matter, white matter beneath. Then add the blood-brain barrier as a note beside the cortex rungs with its three components. Four kinds of protection are on this ladder, so mark each one.

6 **GRID**

Cranial nerve nuclei by brainstem region

Region down the side, midbrain, pons, and medulla. Which cranial nerve nuclei sit there, and one other structure of that region across the top. Then add a top row for the two nerves that attach above the brainstem, and name what they attach to instead.

7 **GRID**

Six lesions and the deficits they name

Lesion site down the side: the internal capsule, the medulla, Broca's area, the jugular foramen, the phrenic nerve, and a CNS tract stripped of myelin. Structure damaged, deficit produced, and the chapter it came from across the top. Fill it from memory before you check it.

8 **CHAIN**

CSF, choroid plexus back to the blood

Eight boxes with arrows, from the lateral ventricles to the arachnoid granulations. This is the second time you have drawn this route, so do it with the packet closed and time yourself. Under thirty seconds means you have it.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

The brain, from memory, nothing open

BRAIN DUMP FIRST

Three minutes. Every region of the brain and brainstem, and the fluid spaces inside them.

Draw the brain in midsagittal section from memory alone. Put in the cerebrum with the corpus callosum and the cingulate gyrus, the thalamus, hypothalamus, and pineal gland around the third ventricle, the midbrain with its colliculi and aqueduct, the pons, the medulla, and the cerebellum with the arbor vitae and the fourth ventricle in front of it. Then add the falx cerebri and the tentorium cerebelli in position and mark the superior sagittal sinus.

In your notebook, head the page **M5-13 The brain, from memory, nothing open** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can complete this drawing without opening anything, and every correction you then make in a second color is a named gap rather than a vague feeling.

DRAWING 2

A spinal cord segment with everything attached to it

BRAIN DUMP FIRST

Two minutes. Cord, coverings, roots, nerve, and the autonomic chain beside it.

Draw one cord segment in cross-section inside a vertebra. Add the gray horns and white columns, then wrap it in pia, arachnoid with the subarachnoid space, and dura, with the epidural space outside that. Bring out the rootlets, both roots, the posterior root ganglion, and the mixed spinal nerve, then split it into all four branches. Finally put a sympathetic trunk ganglion beside the vertebral body and connect it with the rami communicantes.

In your notebook, head the page **M5-14 A spinal cord segment with everything attached to it** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name every structure a needle would pass through from the skin of the back to the central canal, using only this drawing.

DRAWING 3

The somatic and autonomic motor pathways, side by side

BRAIN DUMP FIRST

Two minutes. Both routes drawn from the CNS to the effector, with every difference visible.

Draw the cord twice. On the left run the somatic pathway: one motor neuron from the anterior horn, out the anterior root, along the spinal nerve, to a skeletal muscle. On the right run the autonomic pathway: a preganglionic neuron from the lateral horn out to a ganglion, a synapse, then a postganglionic neuron to smooth muscle, cardiac muscle, or a gland. Mark which axons are myelinated, and label the effect each pathway can have on its effector.

In your notebook, head the page **M5-15 The somatic and autonomic motor pathways, side by side** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at the one feature on the right-hand drawing that the left-hand drawing does not have at all, and say what follows from it.

Apply it

3 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient suddenly develops weakness of the right arm and right leg and speaks in short, effortful phrases, though comprehension is intact. Name the hemisphere affected, the artery most likely blocked, and explain both.

Answer. The left hemisphere, and the left middle cerebral artery.

Reasoning. Start with the side, because the crossing settles it. Most corticospinal fibers decussate at the pyramids of the medulla, so a hemisphere controls the opposite half of the body, and right-sided weakness points to the left hemisphere. Now use the second deficit to locate the lesion within that hemisphere: speech production is lost while comprehension is intact, which is Broca's area in the inferior frontal gyrus, usually on the left. One artery supplies both the lateral motor cortex for the body and that frontal speech area, and it is the middle cerebral artery, the largest branch of the internal carotid. The anterior cerebral artery is ruled out because it feeds the medial surface, and the posterior cerebral artery is ruled out because it feeds the occipital lobe and vision is unaffected. The Circle of Willis could not rescue the territory here because a sudden blockage gives the anastomosis no time to open.

Set A · Name it

1. The only root of a spinal nerve that carries a ganglion is the ________________.
2. The glial cell that forms myelin in the CNS is the ________________.
3. The crossover of motor fibers on the front of the medulla is the ________________.
4. The enlarged subarachnoid space below the end of the spinal cord is the ________________.
5. Name the four ways the central nervous system is protected.

   ____

Set B · Predict it

1. A high cervical cord injury occurs at the C3 level. Predict the effect on breathing and name the nerve and plexus responsible.
2. A lesion sits in the lateral gray horn of the T4 segment. Predict which kind of neuron is lost and which division of the nervous system is affected.
3. Myelin is stripped from axons in the white columns of the cord. Predict the effect on signals travelling to and from the brain, and say why the same injury in a peripheral nerve behaves differently.
4. A blockage forms at the median aperture of the fourth ventricle. Predict where CSF backs up and where the fluid still reaches normally.

Set C · Tell it apart

1. Trace a signal from a receptor in the fingertip to the cortex and back out to the muscle, naming every structure it passes through.

2. Explain why a lesion of the anterior horn and a lesion of the posterior root ganglion produce opposite deficits.

3. Compare the location of a somatic motor neuron cell body with the locations of the two autonomic motor neuron cell bodies, and explain what the ganglion adds.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M5-

The whole nervous system on one page

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M5-2

One stimulus, all the way through

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

M5-3

CNS against PNS, every term that changes

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M5-4

Where every cell body sits

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5–5

From the scalp inward to the neuron

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

M5–6

Cranial nerve nuclei by brainstem region

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5-7

Six lesions and the deficits they name

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M5-8

CSF, choroid plexus back to the blood

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.